



Lab 4: Loading Effect in a Voltage Divider Circuit

Objective:

- To analyze how the voltage divider circuit behaves when there is no load resistance connected.
- Evaluate the performance of voltage divider circuit due to loading.

Theory:

The loading effect in a voltage divider circuit occurs when a load resistance is connected to the output of the divider, causing the output voltage to be lower than the expected value based on the resistor ratios alone. This happens because the load resistor draws current from the divider, effectively reducing the resistance seen by the voltage source and altering the voltage division.

Voltage Divider Basics:

Voltage Divider circuit provides a simple way to convert a DC voltage to another lower DC voltage. Consider the following voltage divider circuit.

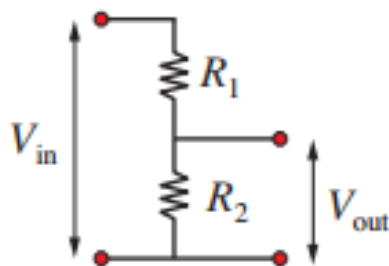


Figure 1: (a) A Voltage divider,

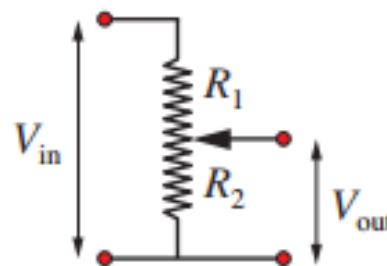


Figure 1: (b) A Potentiometer

The voltage drop across R_2 is the output voltage, V_{out} . V_{out} is less than V_{in} because the total voltage across R_1 and R_2 must add up to V_{in} . A potentiometer can also be used to change V_{out} by changing the resistance R_2 . As the value of R_2 is changed, it allows the output voltage to be adjusted from 0 to V_{in} .

In Figure 1, there is no output load (R_L) connected in parallel to R_2 hence we call it a No-Load circuit.

According to Voltage Divider Rule:

$$V_{out} = V_{R2} = V_{in} \times \frac{R_2}{R_1 + R_2} \dots \dots \dots (1)$$

Now, Say $V_{in}=5V$, and you need $V_{out}= 3V$. How would you set the values of R_1 and R_2 ?

$$\frac{V_{out}}{V_{in}} = \frac{R_2}{R_1 + R_2} = \frac{3}{5}$$

Choice of resistor value should follow the ratio: $\frac{R_1}{R_2} = \frac{2}{3}$

One possible combination: $R_1 = 2k\Omega$ and $R_2 = 3k\Omega$

Loading Effect:

Now say we connect an output load, R_L in parallel to R_2 :

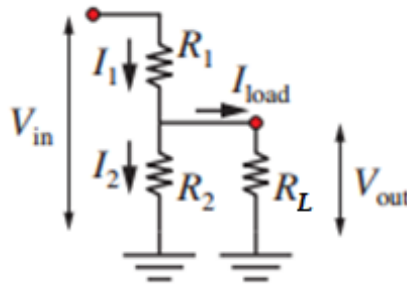


Figure 2: With Output Load Connected

Do you think keeping the values of all resistors same would still give same $V_{out} = 3V$ for same $V_{in} = 5V$?
Let's check:

Since you have a Load resistance R_L in parallel to R_2 , the equivalent resistance will be:

$$R_{eq} = R_2 || R_L = \frac{R_2 R_L}{R_2 + R_L}$$

Using the Voltage divider formula, V_{out} is:

$$V_{out} = V_{in} \times \frac{R_{eq}}{R_{eq} + R_1} \dots \dots \dots (2)$$

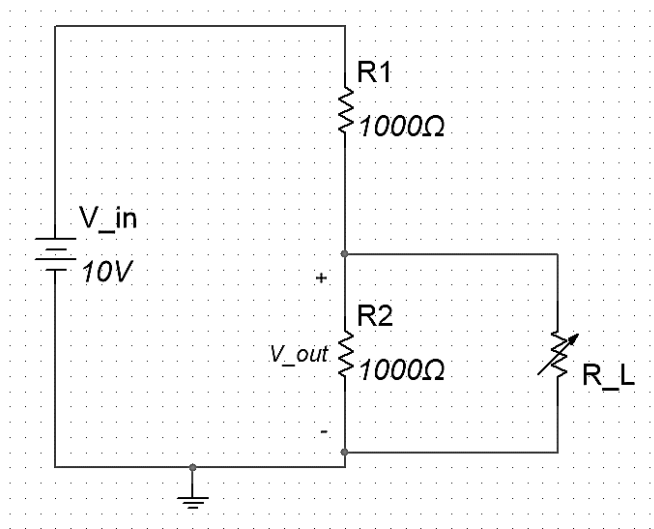
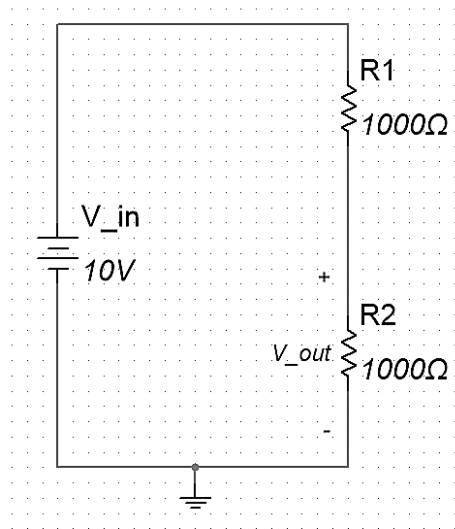
Let $R_L = 10 k\Omega$

So $R_2 || R_L = 2.31 k\Omega$

Then from eqn (2) $V_{out} = 2.68 V$

So, initially, our designed value was 3V, but connecting a load resistor reduced the output voltage to 2.68V.

An Example of Loading Effect:



V_{in}	R_1	R_2	R_L	R_{eq}	$V_{out} = V_{R2}$	I_{R2}
10 V	1000 Ω	1000 Ω	Not Connected	1000 Ω	5 V	5 mA
		1000 Ω	100 Ω	91 $\Omega \downarrow$	0.84 V $\downarrow$	0.84 mA $\downarrow$
		1000 Ω	1000 Ω	500 Ω	3.35 V	3.35 mA
		1000 Ω	2000 Ω	667 Ω	4.0 V	4 mA
		1000 Ω	12000 Ω	923 Ω	4.8 V	4.8 mA

Since the load R_L forms a parallel path with R_2 , the effective resistance R_{eq} decreases, leading to higher total current flow and a lower output voltage. This is the essence of the loading effect.

Summary:

(a) In an Unloaded Voltage Divider (No Load Connected)

- The current flowing through the resistors is determined only by R_1 and R_2 .
- This current remains stable and predictable.

(b) When a Load is Connected (Loaded Circuit)

- The load resistor R_L creates an extra pathway for current to flow.
- This causes more current to flow through the voltage divider circuit. So total current at the output terminal will be higher.
- Since current follows Ohm's Law ($I = V/R$), a smaller load resistance means higher current draw by the load, larger load resistance means less current draw by the load. This affects the current I_{R2} .

Design Criteria:

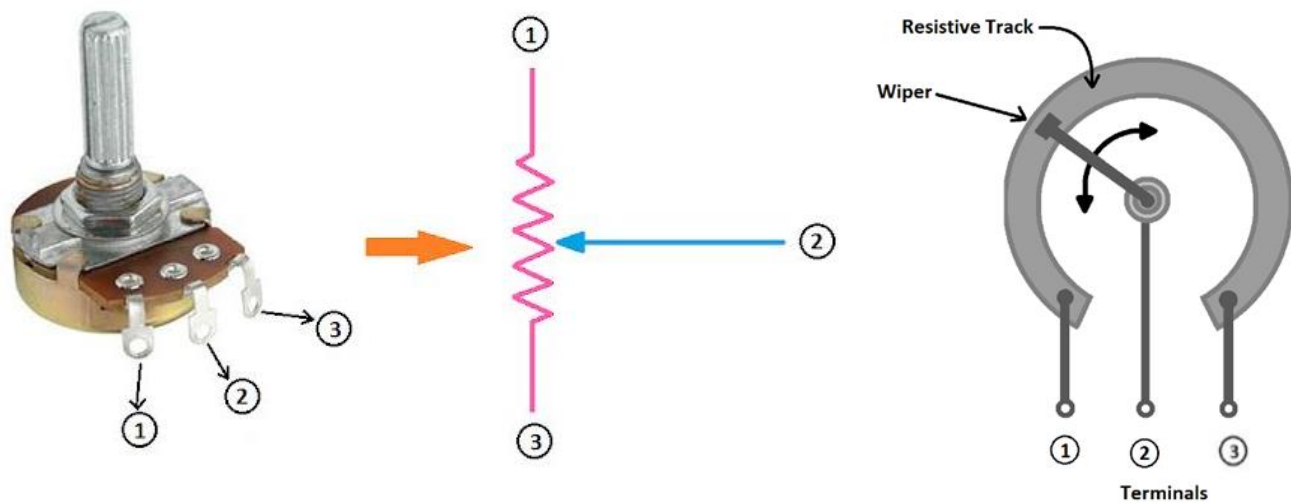
To minimize the loading effect, choose the load resistor to be much larger than its parallel resistor (at least 10 times larger). Larger R_{load} means less current is drawn by the load, causing a smaller change in the voltage division.

If the load resistor R_L is much greater than R_2 , then $R_2 || R_L$ (parallel combination of R_2 and R_L) is approximately equal to R_2 .

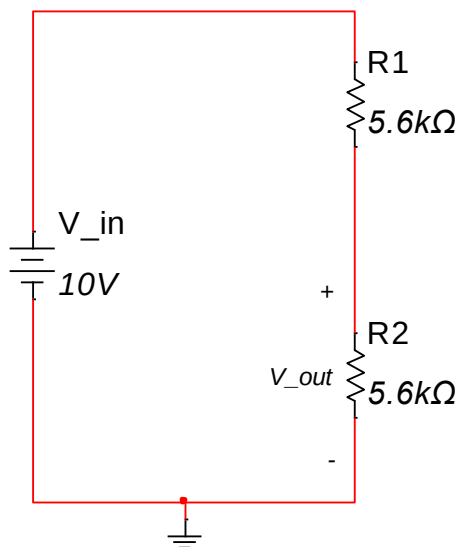
List of Components:

Component Name	Value (or Model Number)	Quantity
Breadboard		1
DC Power Supply (Variable)		1
Resistors	5.6 k Ω	2
Variable Resistor / Potentiometer	10 k Ω POT	1
	100 k Ω POT	1
Connecting Wires		10
Digital Multimeter (DMM)		1

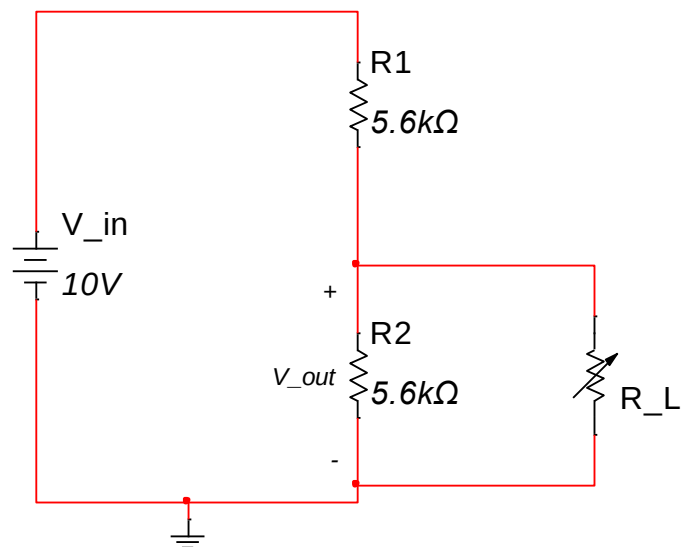
Variable Resistor / Potentiometer:



Circuit Diagram:



Circuit 1



Circuit 2

Procedure:

1. Construct the voltage divider circuit 1 as shown in figure above.
2. Measure the unloaded output voltage V_{out} . Record the value in Table 1 on **No load resistor** row.
3. Connect the 10 k Ω (and then 100 k Ω) variable load resistor, parallel with R_2 to the circuit as shown in circuit 2. (Connect 1 middle pin of variable resistor and one of the other pins).
4. Change the value of the variable resistor according to Table 1, and record V_{out} in Table 1.

Data Collection:

Lab 4:

Group No. _____

Instructor's Signature _____

Measure: R1 = _____

R2 = _____

Table 1: To check the effect on voltages

R_L	R_L (Measured)	V_{R1} (Measured)	$V_{R2} = V_{R_L} = V_{out}$ (Measured)	$V_{R2} = V_{R_L} = V_{out}$ (Calculated)	% of Error in V_{out}
No load resistor	X				
1k Ω					
3.3k Ω					
5k Ω					
10k Ω					
22k Ω					
33k Ω					
68k Ω					
90k Ω					

Table 2: To check the effect on current flow

R_L	R_L (Measured)	I_{R1}	I_{R2}	I_{R_L}
No load resistor	X			X
1k Ω				
3.3k Ω				
5k Ω				
10k Ω				
22k Ω				
33k Ω				
68k Ω				
90k Ω				

Questions:

1. Explain the loading effect of your circuit (i.e explain how does your V_{out} vary with Load Resistor)
2. Showing all steps in details, theoretically calculate the value of V_{out} for each load resistor.
3. Comparing the theoretical data to the experimental data, comment how far the loading effect of your circuit supports the theory.
4. What are the design criteria to minimize the loading effect on a circuit?
5. Explain the working principle of a potentiometer.

MultiSim Simulation

- Simulate the loading effect circuits you did in lab class. Fill the data tables. Verify loading effect.